

# Boris Andrews CV

 |  |  |  |  [boris.andrews@maths.ox.ac.uk](mailto:boris.andrews@maths.ox.ac.uk)  
 [borisandrews.github.io](https://borisandrews.github.io)

## EMPLOYMENT

- 2025 – 2027 **Postdoctoral Research Associate, Numerical Analysis, University of Oxford**  
(predicted)
  - Project: ERC Starting Grant for *Geometric Finite Element Methods (GeoFEM)*
  - Advisor: *Kaibo Hu*

## EDUCATION

- 2021 – 2025 **PhD (DPhil), Numerical Analysis, University of Oxford**  
  - Thesis: *Geometric numerical integration via auxiliary variables*
  - Supervisors: *Patrick Farrell, Wayne Arter*
- 2017 – 2021 **Integrated Masters (MMath), Mathematics, University of Oxford**  
  - Grade: *First (Distinction)*
  - Thesis: *Computation and approximation of near orthogonal matrices for tall random matrices*
  - Supervisor: *Yuji Nakatsukasa*

## RESEARCH INTERESTS

**Structure-preserving/compatible numerical methods for PDEs & ODEs, Conservation & dissipation structures | Global & local energy estimates & conservation laws | Asymptotic-preserving integrators | Geometric machine learning**

**Finite element theory, Finite element exterior calculus (FEEC) | Domain decomposition | Parallel in time (PinT)**

**Plasma modelling, Magnetohydrodynamics (MHD) | Hybrid fluid-particle models**

**Turbulent systems, Stabilisation | Preconditioning**

## TEACHING

- 2026 – 2027 **Lecturer, Mixed Finite Element Methods and Finite Element Complexes, Taught Course Centre** (Bath, Bristol, Imperial, Oxford, Warwick, Swansea)  
  - 16-hour graduate course
- 2024 – 2025 **Tutor, *Computational Mathematics***
- 2023 – 2024 **Tutor, Prelims corner**  
**Teaching assistant, *Numerical Linear Algebra***
- 2021 – 2022 **Teaching assistant, *Random Matrix Theory***  
**Tutor, *Analysis I*, Oriel College**

## SUPERVISION

- Feb – Aug 2026 **Oscar Benton**, MMSC dissertation  
○ Project: *Numerical evolution of magnetic field topologies*
- Feb – Aug 2025 **Matin Shams**, MMSC special topic & dissertation  
○ Project: *Enstrophy-stable integrators for the incompressible Navier–Stokes equations*
- Sep – Oct 2024 **Sebastian Ohlig**, Undergraduate summer internship  
○ Project: *Stability study of conservative vs. symplectic integrators on the Toda lattice*

## PUBLICATIONS & PREPRINTS

### Papers

- Apr 2026 **Helicity-preserving finite element discretization for magnetic relaxation**, with *Mingdong He, Patrick Farrell, Kaibo Hu*  
○ Publication: *SIAM Journal on Scientific Computing (SISC)*, 48 (2), pp. B165–B183
- Dec 2025 **Enforcing conservation laws and dissipation inequalities numerically via auxiliary variables**, with *Patrick Farrell*  
○ Publication: *SIAM Journal on Scientific Computing (SISC)*, 47 (6), pp. A3516–A3535

### Preprints

(In review)

- Jun 2026 **Automated Galerkin time stepping in Irsome**, with *Pablo Brubeck, Patrick Farrell, Robert Kirby, Scott MacLachlan*  
○ In review: *SIAM Journal on Scientific Computing (SISC)*
- May 2026 **Arbitrary-order structure-preserving discretizations for geometric curvature flows**, with *Ganghui Zhang, Patrick Farrell*  
○ In review: *SIAM Journal on Scientific Computing (SISC)*
- Nov 2025 **Conservative and dissipative discretisations of multi-conservative ODEs and GENERIC systems**, with *Patrick Farrell*  
○ In review: *Computers & Mathematics with Applications (CAMWA)*

### Other works

- Jul 2025 **Geometric numerical integration via auxiliary variables**  
○ Note: *PhD (DPhil) thesis*

## TALKS (\*upcoming)

### INVITED TALKS & MINISYMPOSIUM PRESENTATIONS

- 2027 **ICOSAHOM\*** (*Politecnico di Milano*) | **Oberwolfach Workshop on Structure-Preserving Methods\*** (*Oberwolfach Research Institute for Mathematics*)
- 2026 **SciCADE** (*University of Edinburgh*) | **ECCOMAS World Congress on Computational Mechanics\*** (*Munich, Germany*) | **Invited Seminar** (*Chinese Academy of Sciences*)
- 2025 **(2×)ACOMEN** (*Ghent University*) | **ECCOMAS Thematic Conference on Modern Finite Element Technologies** (*Aachen, Germany*) | **Self-Consistency Group Seminar** (*CHaRMNET*) | **ACM Colloquium** (*University of Edinburgh × Heriot-Watt University*) | **Numerical Mathematics & Scientific Computing Seminar** (*Rice University*) | **SIAM CSE** (*Fort Worth, Texas*) | **Scientific Computing Seminar** (*Brown University*) | **METHODS Group Seminar** (*Brown University*)
- 2024 **External Seminar** (*Rice University*)

### OTHER SEMINAR, WORKSHOP & CONFERENCE PRESENTATIONS

- 2026 **Firedrake User Meeting\*** (*University of Oxford*) | **Programme on Differential Complexes** (*University of Vienna*) | **Workshop on Finite Element Tensor Calculus** (*Tsinghua University*)
- 2025 **Numerical Analysis Group Internal Seminar** (*University of Oxford*) | **Biennial Numerical Analysis Conference** (*University of Strathclyde*)
- 2024 **Computing Division Technical Meeting** (*UKAEA*) | **Firedrake User Meeting** (*University of Oxford*) | **PDEsoft** (*University of Cambridge*) | **European Finite Element Fair** (*University College London*) | **Exploiting Algebraic and Geometric Structure in Time-integration Methods Workshop** (*University of Pisa*) | **UKAEA PhD Student Engagement Day** (*UKAEA*) | **Junior Applied Mathematical Seminar** (*University of Warwick*)
- 2023 **ICIAM** (*Waseda University*) | **Numerical Analysis Group Internal Seminar** (*University of Oxford*) | **Junior Applied Mathematics Seminar** (*University of Oxford*) | **Met Office Presentation** (*University of Oxford*)
- 2022 **PRISM Workshop** (*Missenden Abbey, UK*)

---

## HOSTED MINISYMPOSIA

- Jul 2026 **SciCADE, University of Edinburgh**  
○ Topic: *Topology-preserving finite element methods for magnetohydrodynamics*  
○ Co-organisers: *Patrick Farrell, Kaibo Hu*  
○ Speakers: *Tobias Blickhan, Mingdong He, Gunnar Hornig, Shipeng Mao, Adelle Wright, Anthony Yeates, Qian Zhang*
- Jun 2025 **Biennial Numerical Analysis Conference, University of Strathclyde**  
○ Topic: *Structure-preserving finite element methods*  
○ Co-organiser: *Charlie Parker*  
○ Speakers: *Aaron Brunk, Mingdong He, Kaibo Hu, Rami Masri*

---

## OTHER EXPERIENCE

- Sep – Dec 2026 **ETH Zurich, Co-editor of the Nachdiplom Lectures on Finite Element Tensor Calculus with Kaibo Hu**  
○ Contributed volume in preparation for publication
- 2025 – present **University of Oxford, Organisation of the Numerical Analysis Group's weekly finite element methods reading group**
- Apr – Jun 2026 **University of Vienna, Attendance at the Programme on Differential Complexes at the Erwin Schrödinger International Institute for Mathematics and Physics (ESI)**
- Aug – Oct 2022 **Tokamak Energy, Internship, Physics: theory and modelling**  
○ Project: *Implementation of non-Maxwellian backgrounds in the GENE gyrokinetic code*  
○ Supervisor: *Salomon Janhunnen*
- Jul 2022 **United Kingdom Atomic Energy Authority (UKAEA), Plasma physics summer school**

---

## LANGUAGES

### Programming

**Experienced:** Python (*Firedrake*), MATLAB, LaTeX | **Limited:** Julia, C, Fortran

### Spoken

**Fluent:** English | **Intermediate:** Dutch | **Beginner:** Japanese, German